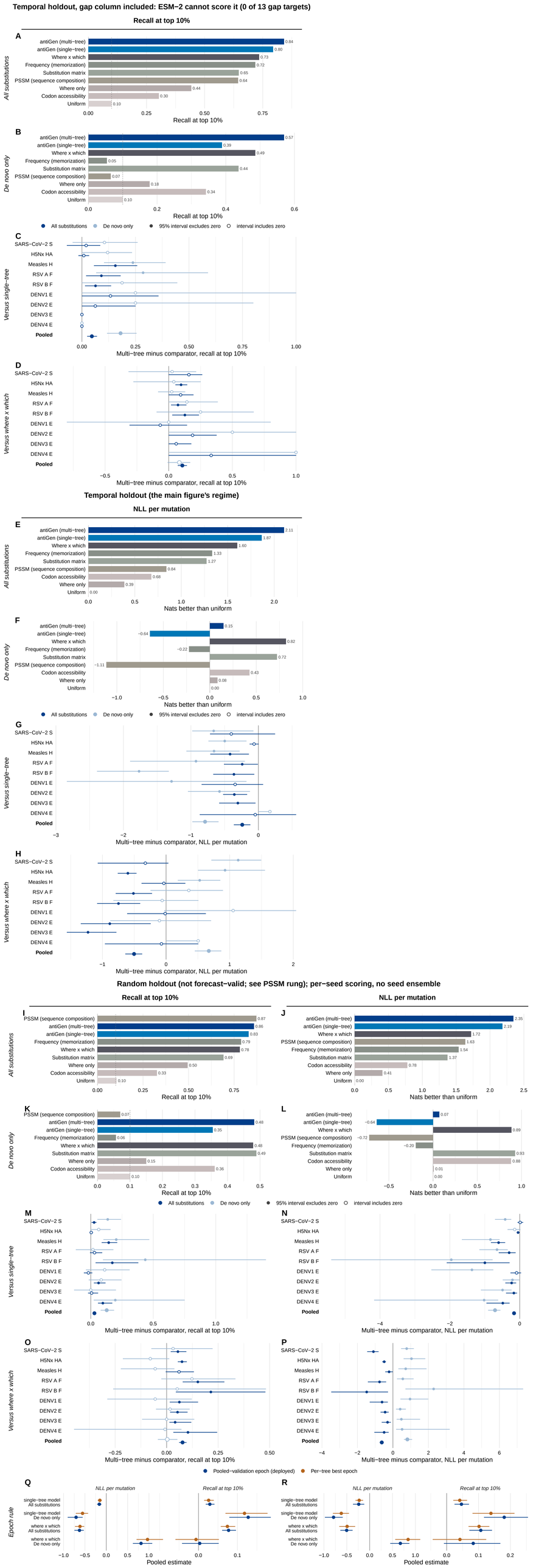
Temporal holdout, gap column included: ESM−2 cannot score it (0 of 13 gap targets)
Recall at top 10%
A
All substitutions
antiGen (multi−tree) 0.84
antiGen (single−tree) 0.80
Where x which 0.73
Frequency (memorization) 0.72
Substitution matrix 0.65
PSSM (sequence composition) 0.64
Where only 0.44
Codon accessibility 0.30
Uniform 0.10
0.00 0.25 0.50 0.75
Recall at top 10%
B
De novo only
antiGen (multi−tree) 0.57
antiGen (single−tree) 0.39
Where x which 0.49
Frequency (memorization) 0.05
Substitution matrix 0.44
PSSM (sequence composition) 0.07
Where only 0.18
Codon accessibility 0.34
Uniform 0.10
0.0 0.2 0.4 0.6
Recall at top 10%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
C
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
0.00 0.25 0.50 0.75 1.00
Multi−tree minus comparator, recall at top 10%
D
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−0.5 0.0 0.5 1.0
Multi−tree minus comparator, recall at top 10%
Temporal holdout (the main figure's regime)
NLL per mutation
E
All substitutions
antiGen (multi−tree) 2.11
antiGen (single−tree) 1.87
Where x which 1.60
Frequency (memorization) 1.33
Substitution matrix 1.27
PSSM (sequence composition) 0.84
Codon accessibility 0.68
Where only 0.39
Uniform 0.00
0.0 0.5 1.0 1.5 2.0
Nats better than uniform
F
De novo only
antiGen (multi−tree) 0.15
antiGen (single−tree) −0.64
Where x which 0.82
Frequency (memorization) −0.22
Substitution matrix 0.72
PSSM (sequence composition) −1.11
Codon accessibility 0.43
Where only 0.08
Uniform 0.00
−1.0 −0.5 0.0 0.5 1.0
Nats better than uniform
All substitutions
De novo only
95% interval excludes zero
interval includes zero
G
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−3 −2 −1 0
Multi−tree minus comparator, NLL per mutation
H
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−1 0 1 2
Multi−tree minus comparator, NLL per mutation
Random holdout (not forecast−valid; see PSSM rung); per−seed scoring, no seed ensemble
Recall at top 10%
NLL per mutation
I
All substitutions
PSSM (sequence composition) 0.87
antiGen (multi−tree) 0.86
antiGen (single−tree) 0.83
Frequency (memorization) 0.79
Where x which 0.78
Substitution matrix 0.69
Where only 0.50
Codon accessibility 0.33
Uniform 0.10
0.00 0.25 0.50 0.75
Recall at top 10%
J
antiGen (multi−tree) 2.35
antiGen (single−tree) 2.19
Where x which 1.72
PSSM (sequence composition) 1.63
Frequency (memorization) 1.54
Substitution matrix 1.37
Codon accessibility 0.78
Where only 0.41
Uniform 0.00
0.0 0.5 1.0 1.5 2.0 2.5
Nats better than uniform
K
De novo only
PSSM (sequence composition) 0.07
antiGen (multi−tree) 0.48
antiGen (single−tree) 0.35
Frequency (memorization) 0.06
Where x which 0.48
Substitution matrix 0.49
Where only 0.15
Codon accessibility 0.36
Uniform 0.10
0.0 0.1 0.2 0.3 0.4 0.5
Recall at top 10%
L
antiGen (multi−tree) 0.07
antiGen (single−tree) −0.64
Where x which 0.89
PSSM (sequence composition) −0.72
Frequency (memorization) −0.20
Substitution matrix 0.93
Codon accessibility 0.88
Where only 0.01
Uniform 0.00
−0.5 0.0 0.5 1.0
Nats better than uniform
All substitutions
De novo only
95% interval excludes zero
interval includes zero
M
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
0.0 0.5 1.0
Multi−tree minus comparator, recall at top 10%
N
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−4 −2 0
Multi−tree minus comparator, NLL per mutation
O
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−0.25 0.00 0.25 0.50
Multi−tree minus comparator, recall at top 10%
P
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−4 −2 0 2 4 6
Multi−tree minus comparator, NLL per mutation
Pooled−validation epoch (deployed)
Per−tree best epoch
Q
Epoch rule
NLL per mutation
Recall at top 10%
single−tree model All substitutions
single−tree model De novo only
where x which All substitutions
where x which De novo only
−1.0 −0.5 0.0 0.5 1.0 0.0 0.1
Pooled estimate
R
NLL per mutation
Recall at top 10%
single−tree model All substitutions
single−tree model De novo only
where x which All substitutions
where x which De novo only
−1.0 −0.5 0.0 0.5 1.0 0.0 0.1 0.2
Pooled estimate